As the field of machine learning developed, algorithms gained more independence. The fundamental principle of machine learning is that algorithms can teach themselves new things by interacting with the world, just as humans do, thereby producing a fully fledged artificial intelligence. The terminology is not always consistent, but generally speaking, for something to be acknowledged as an AI, it needs the capacity to learn new things by itself, rather than just follow the instructions of its original human creators. Present-day chess-playing AI is taught nothing except the basic rules of the game. It learns everything else by itself, either by analysing databases of prior games or by playing new games and learning from experience.[59] AI is not a dumb automaton that repeats the same movements again and again irrespective of the results. Rather, it is equipped with strong self-correcting mechanisms, which allow it to learn from its own mistakes.

This means that AI begins its life as a 'baby algorithm' that has a lot of potential and computing power but doesn't actually know much. The AI's human parents give it only the capacity to learn and access to a world of data. They then let the baby algorithm explore the world. Like organic newborns, baby algorithms learn by spotting patterns in the data to which they have access. If I touch fire, it hurts. If I cry, Mum comes. If I sacrifice a queen for a pawn, I probably lose the game. By finding patterns in the data, the baby algorithm learns more, including many things that its human parents don't know.[60]

Yet databases come with biases. The face-classification algorithms studied by Joy Buolamwini were trained on data sets of tagged online photos, such as the Labeled Faces in the Wild database. The photos in that database were taken mainly from online news articles. Since white males dominate the news, 78 per cent of the photos in the database were of males, and 84 per cent were of white people. George W. Bush appeared 530 times – more than twice as many times as all Black women combined.[61] Another database prepared by a US government agency was more than 75 per cent male, was almost 80 per cent light-skinned, and had just 4.4 per cent dark-skinned females.[62] No wonder the algorithms trained on such data sets were excellent at identifying white men but lousy at identifying Black women. Something similar happened to the chatbot Tay. The Microsoft engineers didn't build into it any intentional prejudices. But a few hours of